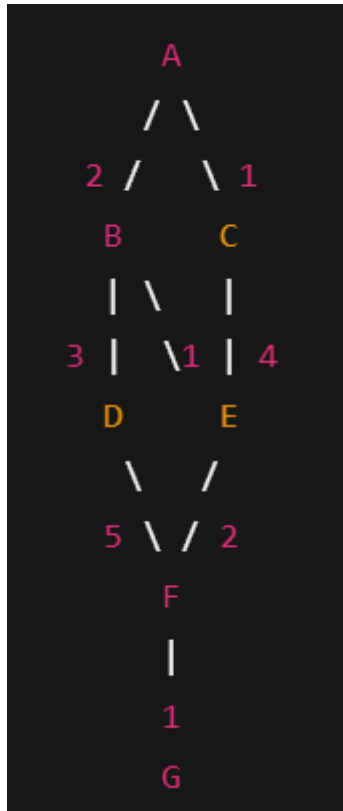


Practical Questions Samples

1. Graph Traversal/Pathfinding



Node	Heuristic h(n)	Meaning
A	7	Estimated cost from A to G
B	6	Estimated cost from B to G
C	4	Estimated cost from C to G
D	5	Estimated cost from D to G
E	3	Estimated cost from E to G
F	2	Estimated cost from F to G
G	0	Goal node (no cost to itself)

Given the following weighted graph, use:

- BFS to find a path from node A to node G
- DFS to find a path from node A to node G
- UCS to find the cheapest path
- A* using a provided heuristic table
- Greedy search using the same heuristic

2. Given the following dataset:

Height (CM)	Weight (KG)	Class
167	51	Underweight
182	62	Normal
176	69	Normal
173	64	Normal
172	65	Normal
174	56	Underweight
169	58	Normal
173	57	Normal
170	55	Normal
170	57	?

- Create this dataset as csv file and solve the following
- Split the data into training set (70%) and testing set (30%) using - train_test_split - from scikit-learn.
- Train the following classifiers:
 - 1) **K-Nearest Neighbors (with n_neighbors=3)**
 - 2) **Gaussian Naive Bayes**
- For each model:
 - 1) Predict the labels on the test set.
 - 2) Report:
 - **Accuracy score.**